



HUMANE Paper Review

LLM Agents Grounded in Self-Reports Enable General-Purpose Simulation of Individuals

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LLM Agents Grounded in Self-Reports Enable General-Purpose Simulation of Individuals

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✓ Predicting Human Behavior: The Bottleneck

Machine learning predicts behavior well — but only where big, structured, outcome-specific data already exist.



Outcome-specific

One trained model per target outcome
— churn, voting, dropout ...



Data-hungry

Each new outcome needs costly, in-domain training data



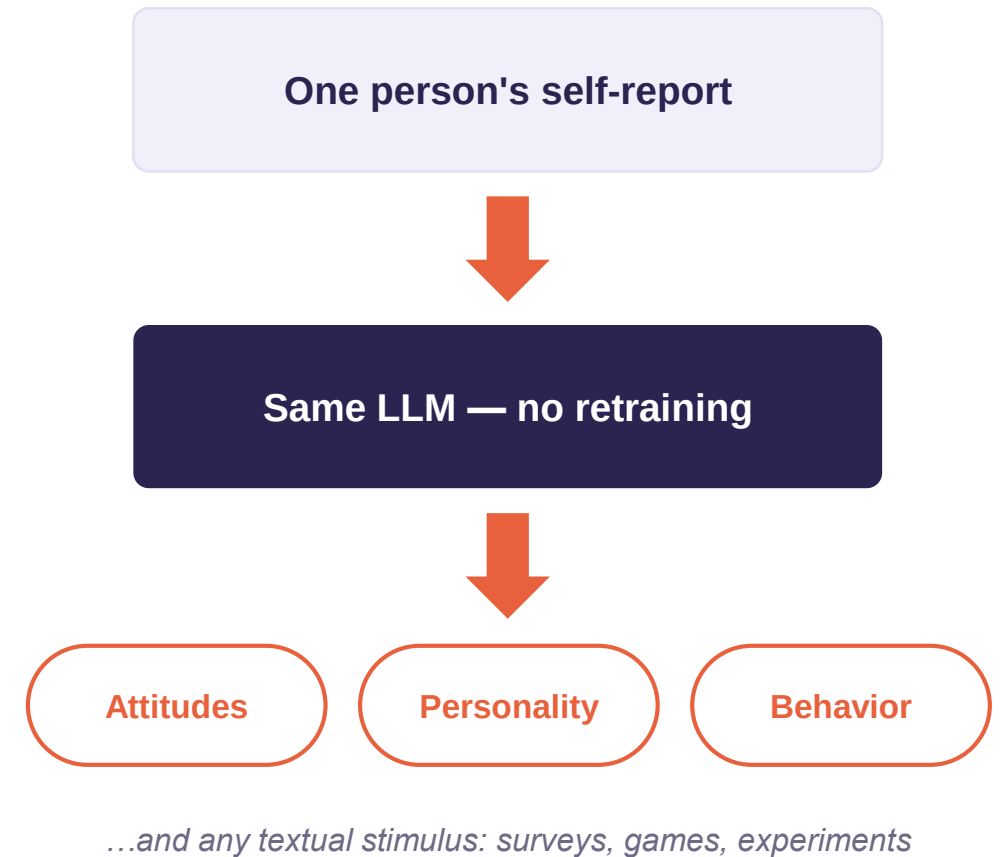
Hard to transfer

Unavailable for many settings scientists and policymakers care about

Background

✓ LLMs as a General-Purpose Interface

- One description of a person
→ the same model can be queried across many outcomes
- No outcome-specific training required
- But prior work grounds agents in demographics
→ stereotyped predictions (refs. 18–20)
- This paper: ground agents in people's own self-reports — interviews & surveys



Research Questions

1. Prediction

- **Can self-report-grounded agents predict held-out individual responses — without task-specific training?**

2. Data source

- **Which grounding works better: in-depth interviews, structured surveys, or both combined?**

3. Fairness

- **Does richer personal data reduce accuracy gaps across demographic groups?**

• Study Design: 1,052 Americans, Two Waves

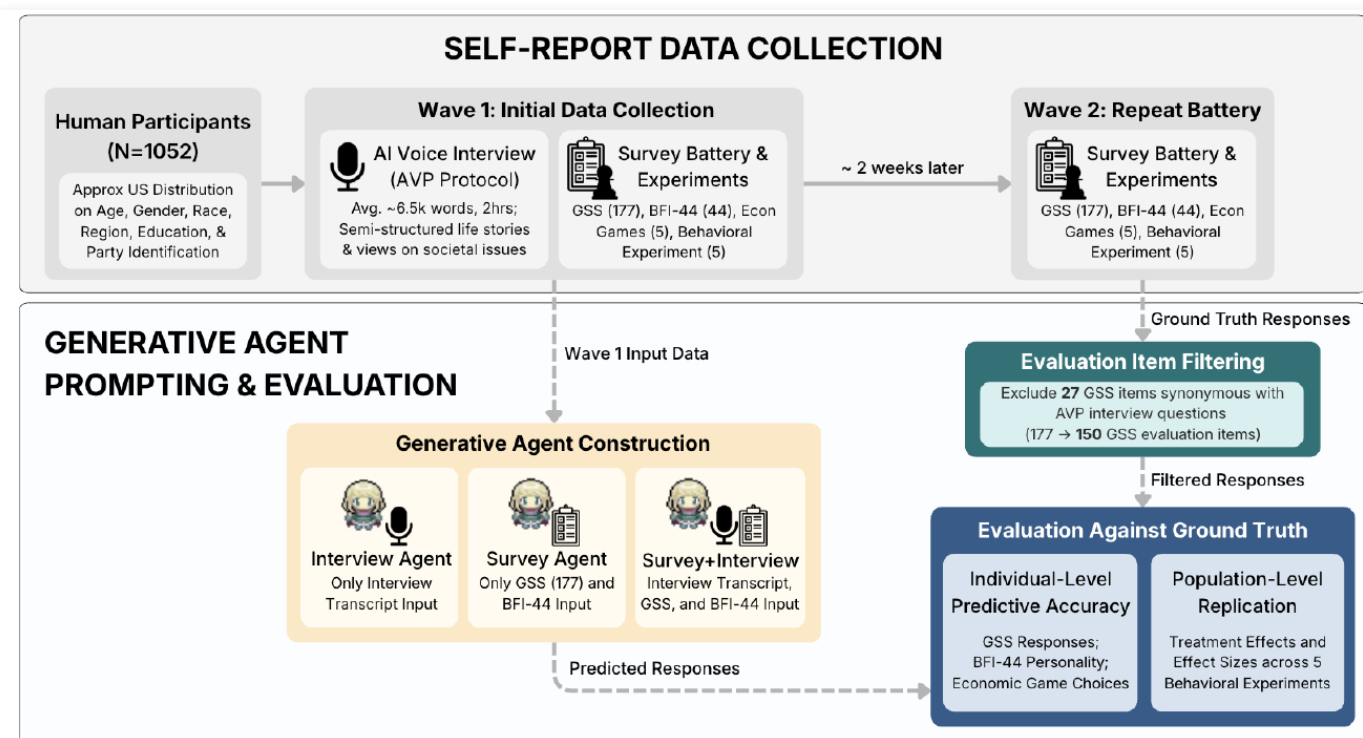


Figure 1. We recruit a stratified U.S. sample of 1,052 individuals by age, gender, race, region, education, and party identification. Participants complete a two-hour audio interview with our AI interviewer, followed by surveys and experiments. For each participant, we create three generative agents: interview agents (transcript only), survey agents (structured responses only), and survey-plus-interview agents (both). Agents and participants complete the same surveys and experiments; participants retake them two weeks later. We assess individual-level accuracy by comparing agent responses to participants' Wave 2 responses, normalized by each participant's self-consistency. Solid arrows denote the chronological progression of human participants across experimental waves; dashed arrows

- Stratified U.S. sample: age, gender, race, region, education, party ID
- Wave 1 — 2-hour AI voice interview + GSS, BFI-44, econ games, experiments
- Wave 2 (+2 weeks) — same battery retaken → test–retest benchmark
- 27 GSS items overlapping the interview removed → 150 evaluation items

✓ **Scaling In-Depth Interviews with an AI Interviewer**

- **Voice-to-voice agent administers the American Voices Project protocol**
- **Protocol designed independently of all evaluation tasks**
- **Reflection module summarizes answers in real time → adaptive follow-up questions**

2 h

average interview length

6,491

words per transcript (avg)

~82

adaptive follow-ups asked

1,052

interviews completed

✓ Five Agent Specifications, One Model (GPT-4o)

SELF-REPORT AGENTS · THIS WORK



Interview agent

Full 2-h interview transcript



Survey agent

GSS + BFI-44 structured responses



Survey + Interview

Both self-report sources combined

BASELINES · PRIOR LITERATURE



Demographic agent

Age, gender, race, political ideology (ref. 2)



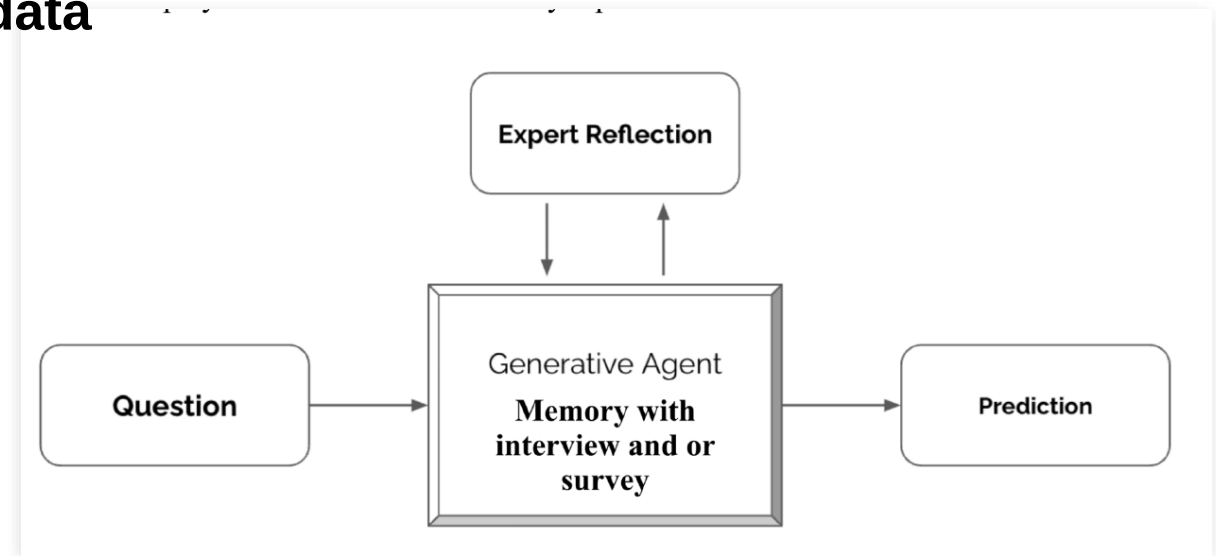
Persona agent

Self-written paragraph about oneself (ref. 39)

Same model, same prompts — only the grounding information differs.

✓ Agent Architecture: Memory + Expert Reflection

- **Memory** = the participant's own self-report data
- **Expert reflection**: 4 personas
 - psychologist, behavioral economist, political scientist, demographer
 - draw out unstated implications
- **At query time**:
relevant expert's notes + chain-of-thought reasoning → prediction



SI Appendix, Figure 4 — generative agent architecture

- **Responds to any textual stimulus (surveys, games, experiment vignettes)**

Evaluation

Normalized accuracy = agent's prediction accuracy $\div$ participant's 2-week test–retest consistency

- **Human answers drift over time $\rightarrow$ raw 100% accuracy is unattainable**
- **1.0 means: "the agent predicts you as well as you replicate yourself two weeks later"**
- **Makes instruments with different reliability comparable**

Example — interview agents on GSS

65.7% $\div$ 79.5%

= 0.83

Evaluation

✓ Four Evaluation Task Families



GSS

150 core items

Attitudes & behaviors across society



BFI-44

Big Five traits

44-item personality inventory



Economic games

5 games, real stakes

Dictator, Trust ×2, Public Goods, PD



Experiments

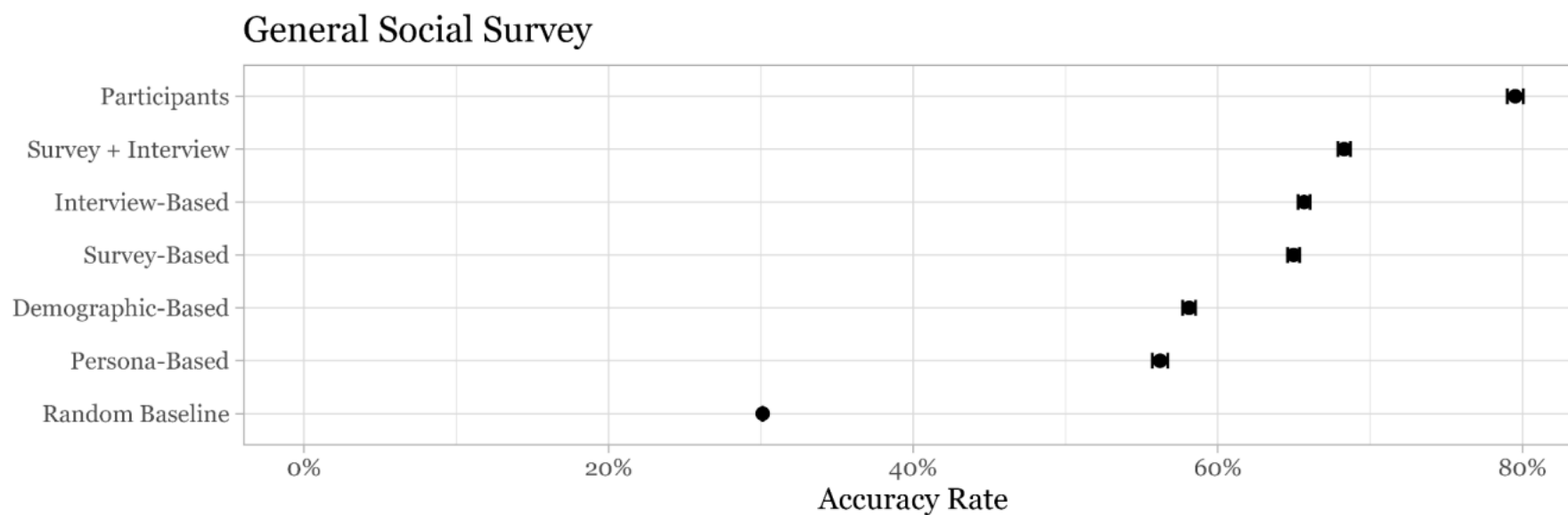
5 replications

Treatment effects (Camerer et al.)

First three → individual-level prediction accuracy · last → population-level replication

Results

✓ GSS: 82–86% of the Test–Retest Ceiling



0.86

Survey + Interview

0.83

Interview

0.82

Survey

0.74

Demographic

0.71

Persona

Results

✓ Big Five: Interviews Carry the Signal

- Interview 0.80 and Survey+Interview 0.77 beat both baselines (normalized correlation, $p < .001$)
- Survey-only agents (0.65) — not distinguishable from demographic (0.61) or persona (0.75) baselines
- Open-ended narratives capture personality cues that structured items miss

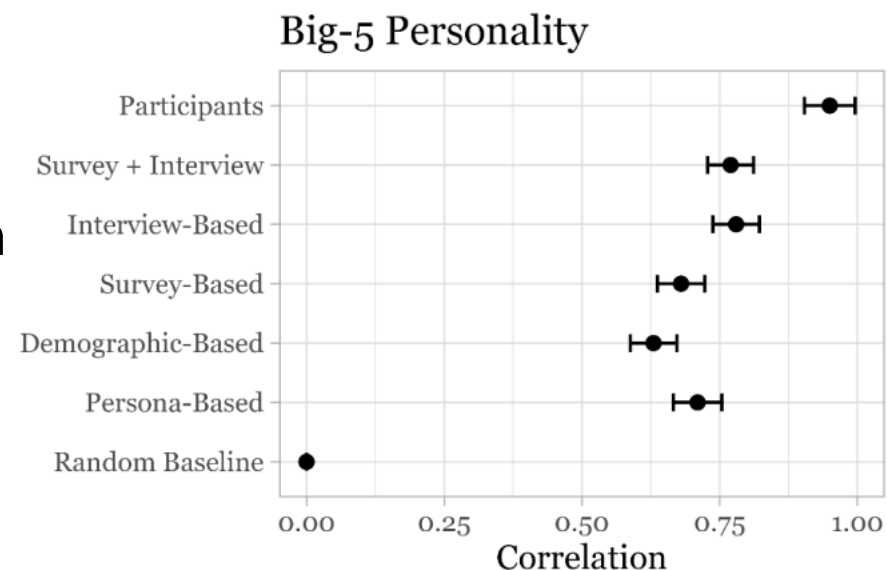


Figure 2 (main paper), bottom-right panel — Big Five correlation

Results

✓ Economic Games: A Boundary Case

- No significant differences across the five agent types — $F(4, 5255) = 1.63$, $p = .16$
- No agent significantly beats the demographic baseline
- Incentivized strategic choices seem to resist self-report grounding
- Authors treat games as a boundary case, not a success domain

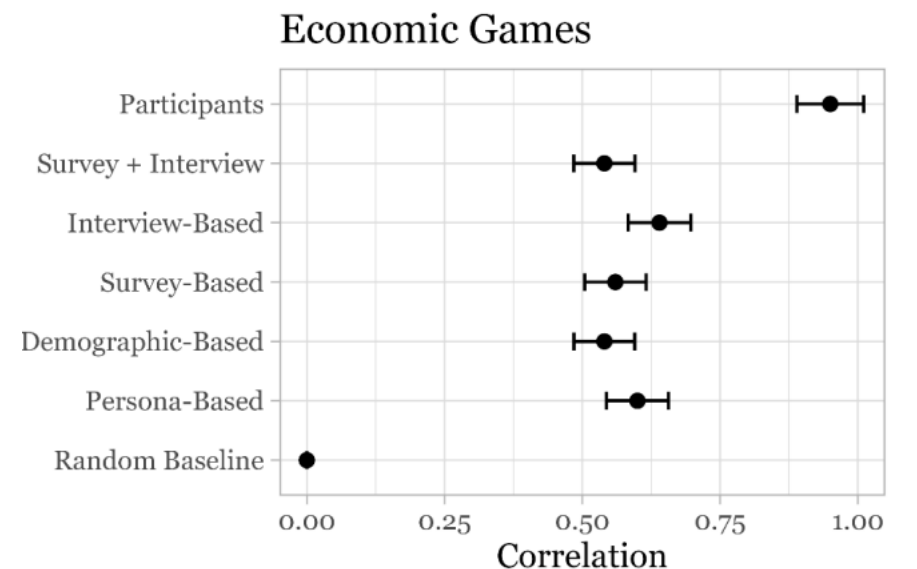


Figure 2 (main paper), bottom-left panel — economic games

Results

✓ Replicating Five Experiments

Replication Studies	Human replication		Agent prediction										
	Participants		Interview		Survey		Survey + interview		Demographic		Persona		
	<i>p</i>	Effect size	<i>p</i>	Effect size	<i>p</i>	Effect size	<i>p</i>	Effect size	<i>p</i>	Effect size	<i>p</i>	Effect size	
Ames & Fiske 2015	***	8.03	***	12.59	***	21.22	***	17.35	***	13.43	***	10.30	
Cooney et al. 2016	***	0.40	***	1.34	***	1.61	***	1.43	***	1.37	***	1.27	
Halevy & Halali 2015	***	0.93	***	2.98	***	3.95	***	3.06	***	4.22	***	3.35	
Rai et al. 2017		0.06		0.094	*	0.16		0.09	***	0.21		0.078	
Schilke et al. 2015	***	0.36	***	2.97	***	9.64	***	3.18	***	5.52	***	3.74	
<i>Effect size correlation w/ human rep.</i>				Correlation		Correlation		Correlation		Correlation		Correlation	
				<i><u>r = 0.98</u></i>		<i><u>r = 0.91</u></i>		<i><u>r = 0.99</u></i>		<i><u>r = 0.93</u></i>		<i><u>r = 0.94</u></i>	
				95% CI [0.75, 1.00]		95% CI [0.15, 0.99]		95% CI [0.87, 1.00]		95% CI [0.26, 1.00]		95% CI [0.38, 1.00]	

$r = 0.91-0.99$

correlation between agent-predicted and human effect sizes, across all agent types

Table 1. Results of replication studies by human participants and generative agents. We report the p-values (***: < 0.001, **: < 0.01, *: < 0.05) and effect sizes.

Direction & relative size match well — but agents inflate effect magnitudes; 5 studies = underpowered

Robustness

✓ How Much Interview Is Enough?

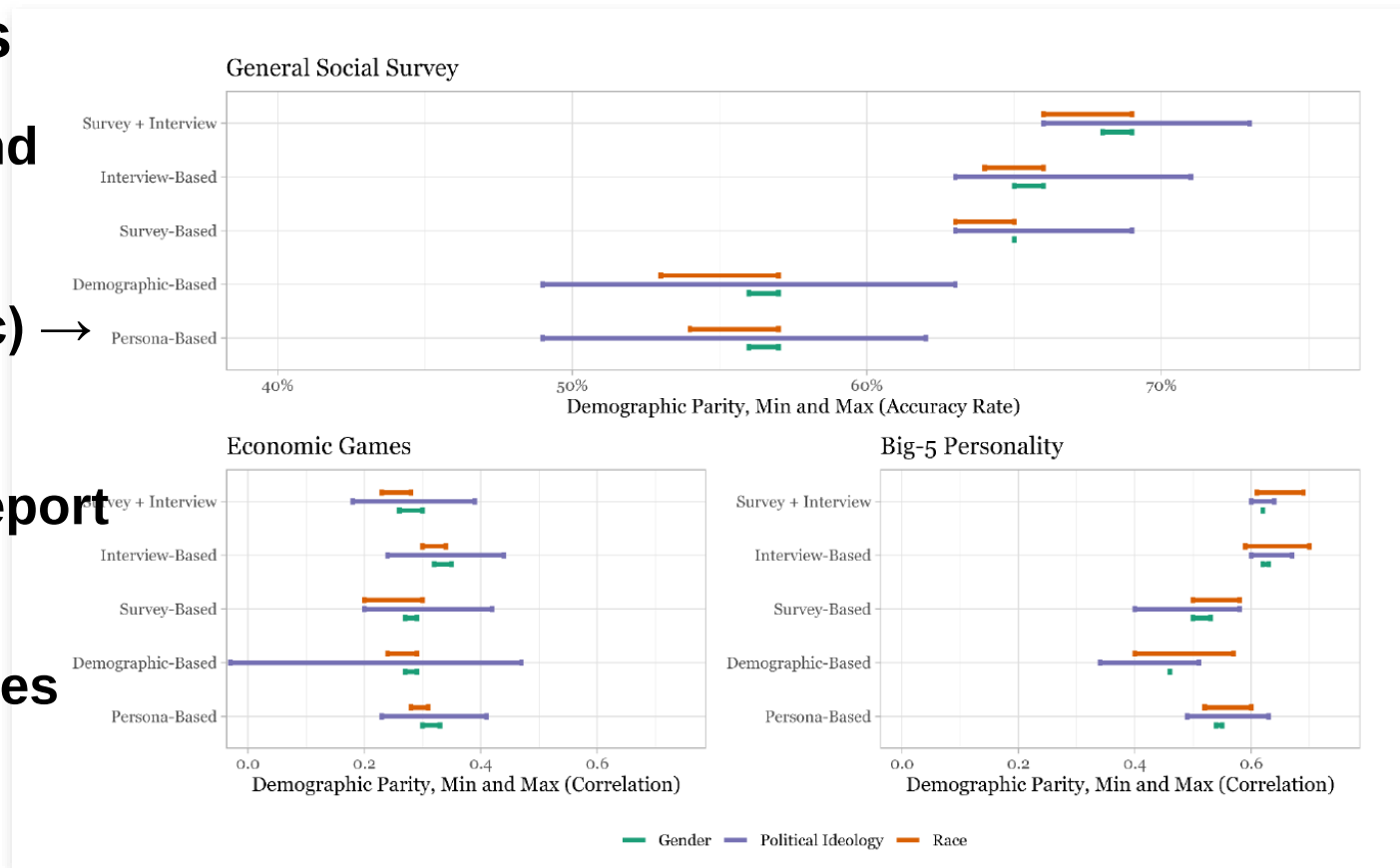
- Randomly delete 80% of the transcript (96 of 120 min) → only 0.82 → 0.79, still above demographic agents
- Fact-only summaries ≈ full transcripts → information content matters, not linguistic style
- Maximal agents (all data): 0.87 — clear diminishing returns

	Five social-science studies
<i>Human replication rate</i>	Accuracy=0.48 (std=0.14) Correl.=0.61 (z std=0.34)
<i>Interview</i>	Acc.=0.24 (std =0.15) Nrm. Acc.=0.50 (std=0.12) -- Correl.=0.34 (std=0.16) Nrm. Correl.=0.56 (std=0.28) -- Correlation with treatment effects: 0.93
<i>Survey Agents</i> (GSS+Big5)	Acc.=0.24 (std = 0.15) Nrm. Acc.=0.52 (std = 0.11) -- Correl.=0.33 (std = 0.14) Nrm. Correl.=0.38 (std = 0.25) -- Correlation with treatment effects: 0.91
<i>Survey agents with interview</i> (GSS+big5+interview)	Acc.=0.25 (std = 0.16) Nrm. Acc.=0.53 (std = 0.12) -- Correl.=0.34 (std = 0.15) Nrm. Correl.=0.38 (std = 0.29) -- Correlation with treatment effects: 0.99
<i>Demographic</i>	Acc.=0.21 (std =0.16) Nrm. Acc.=0.45 (std=0.14) -- Correl.=0.31 (std=0.17) Nrm. Correl.=0.32 (std=0.34) -- Correlation with treatment effects: 0.93
<i>Persona</i>	Acc.=0.23 (std =0.16) Nrm. Acc.=0.48 (std=0.14) -- Correl.=0.33 (std=0.17) Nrm. Correl.=0.54 (std=0.35) -- Correlation with treatment effects: 0.94

Results

✓ Richer Data Also Reduces Bias

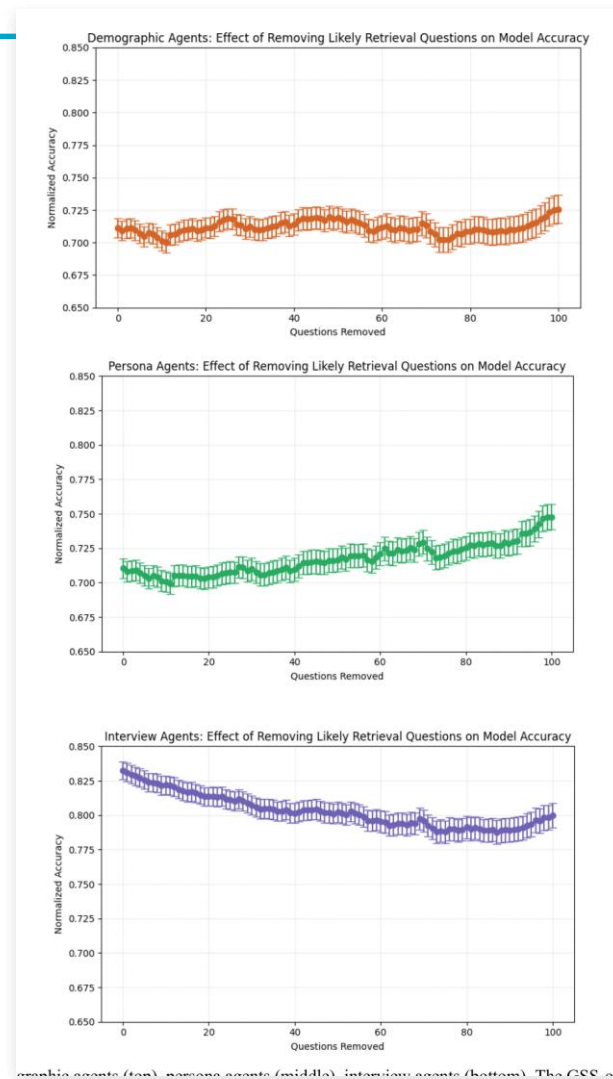
- **DPD = accuracy gap between best- and worst-predicted group**
- **Ideology (GSS): 13.75% (demographic) → 8.60% interview, 6.22% survey**
- **Race (GSS): 3.39% → ~2% with self-report agents**
- **Gender gaps already < 1% — disparities narrowed, not eliminated**



Why does it work?

✓ Mechanism 1 — Direct Retrieval

- Narratives often already contain the answer: “My health is so bad I can't even work” → employment items
- Test: drop GSS items most likely answerable verbatim from the transcript, one by one
- Interview agents lose accuracy as those items are removed
- Transcript-free baselines stay flat → retrieval genuinely contributes



SI Appendix, Figure 7

Why does it work?

✓ Mechanism 2 — Inference

- **Model combines transcript + world knowledge: “enrolled in school” → unlikely to have a workplace supervisor**
- **~3× more GSS items flagged as inferable than as directly retrievable**
- **Same removal test → again, only interview agents decline (baselines flat)**
- **Removing both kinds: advantage shrinks (0.095 → 0.080) but persists → further mechanisms remain unexplained**

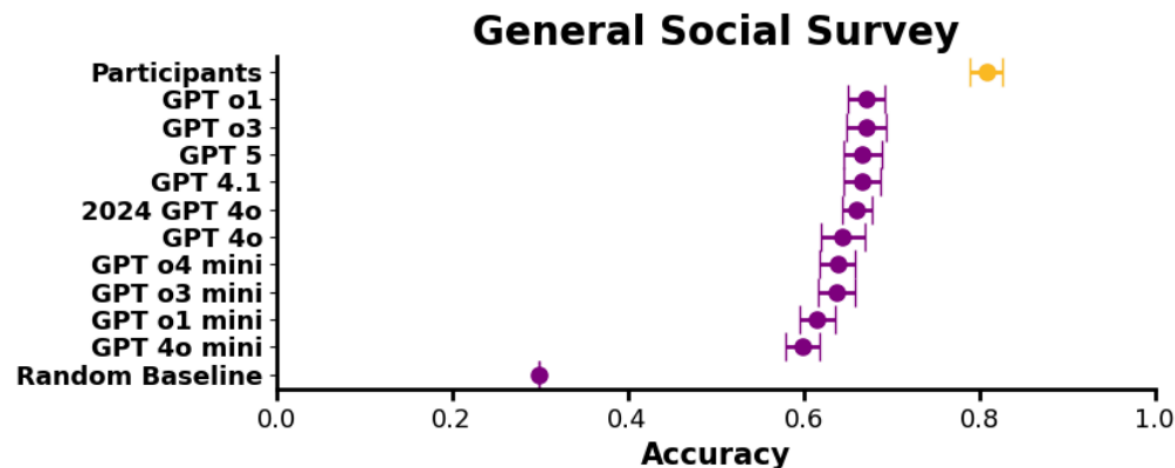


SI Appendix, Figure 8

Robustness

✓ Stable Across Models and Time

- GPT-5 / 4.1 / o1 / o3 ≈ 0.67 vs GPT-4o 0.66 — no significant gain from stronger models
- 2025 rerun of GPT-4o reproduces the 2024 result almost exactly
- Fine-tuning on GSS answers hurts rather than helps (0.79 vs 0.84)
- → the bottleneck is the data, not the model



SI Appendix, Figure 8

Implications for Social Science

- **Pilot surveys and experimental designs before fielding them**
- **Explore treatment-effect heterogeneity before costly fieldwork**
- **Hybrid designs: combine simulated + human data to boost power (ref. 22)**
- **Ethics built in: informed consent, pseudonymization, 25-year withdrawal right**

“The quality of individual simulation depends far less on model scale or persona engineering than on the depth and reliability of the data an agent is built from.”

Takeaways

1. Grounding works

- **Self-report agents recover 82–86% of each person's own test–retest ceiling on the GSS**

2. It generalizes

- **One agent predicts attitudes, personality, and experimental responses — no task-specific training (games remain a boundary case)**

3. It's fairer

- **Richer personal data narrows accuracy gaps across racial and ideological groups**

Strong & Weak

- **Strong**

- Rich self-reports enable more accurate human simulation than demographic prompts
- One grounded agent generalizes across multiple tasks
- Large-scale and rigorous evaluation supports the findings

- **Weak**

- **Mechanism only partly understood — retrieval + inference don't explain everything**
- **Replication analysis: only 5 experiments — underpowered**
- **English-language, U.S. sample only — cross-cultural validity open**
- **Individual accuracy $\neq$ population-level correlational structure of attitudes**
- **Single model family (GPT); transfer to open-weight models untested**